Engine Performance Data

Cummins Inc

Columbus, Indiana 47202-3005
http://www.cummins.com

Industrial

QSL9

FR93996

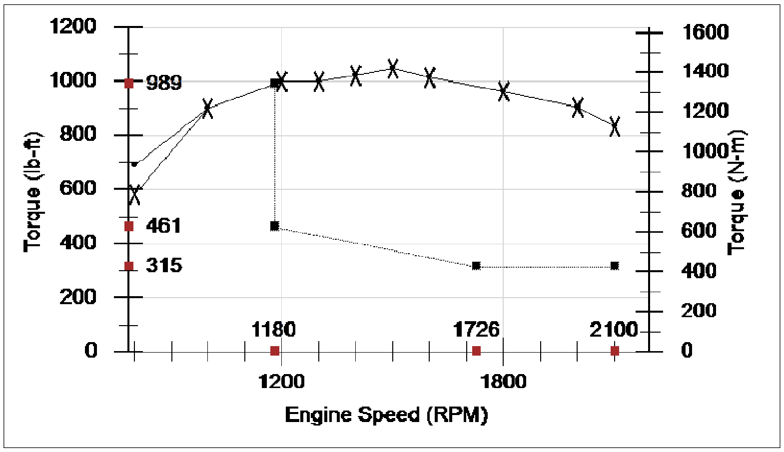
333 BHP (248 kW) @ 2100 RPM
1,050 lb-ft (1,424 N-m) @ 1500 RPM

Configuration	CPL Code	Revision
D563019CX03	**3823-93996**	**5-May-2015**

Compression Ratio:	**16.72:1**	Displacement:	**543 in3 (8.9 L)**
Fuel System:	**XPI**	Aspiration:	**Turbocharged and Charge Air Cooled**
Emission Certification:	**CARB Tier 4(f), EU Stage IV, U.S. EPA Tier 4(f)**		

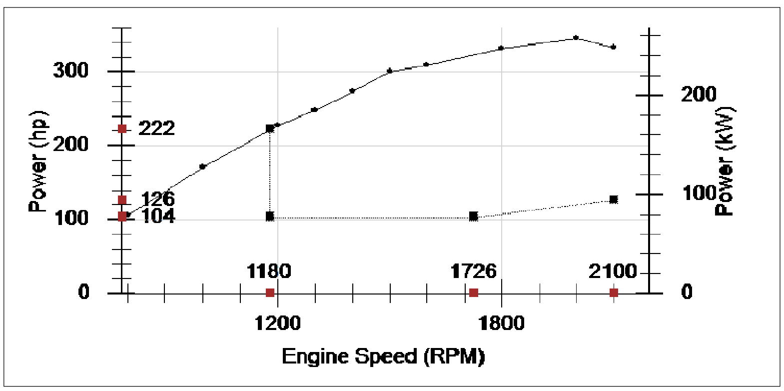
All data is based on the engine operating with fuel system, water pump, and 15 in H2O (3.73) inlet air restriction with 4 in (102 mm) inner diameter, and with 10.1 in Hg (34) exhaust restriction with 3 in (76 mm) inner diameter; not included are alternator, fan, optional equipment and driven components. Coolant flows and heat rejection data based on coolants as 50% ethylene glycol/50% water. All data is subject to change without notice.

Rating Type: Intermittent



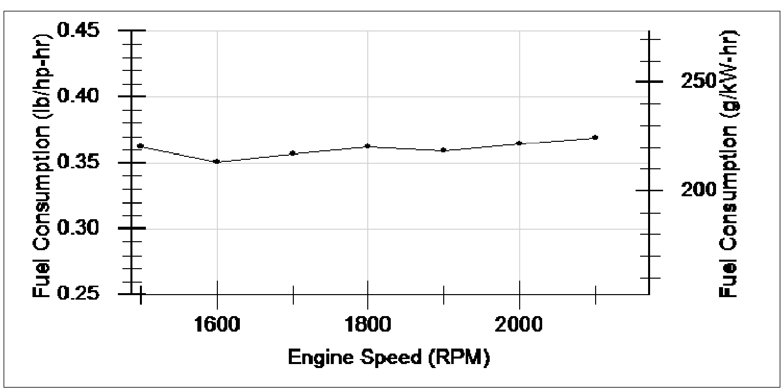
Torque Output

RPM	300 ft lb-ft	300 ft N-m	5,500 lb-ft	5,500 N-m
800	690	936	582	789
1,000	900	1,220	900	1,220
1,200	1,000	1,356	1,000	1,356
1,300	1,000	1,356	1,000	1,356
1,400	1,025	1,390	1,025	1,390
1,500	1,050	1,424	1,050	1,424
1,600	1,015	1,376	1,015	1,376
1,800	965	1,308	965	1,308
2,000	905	1,227	905	1,227
2,100	834	1,131	834	1,131



Power Output

RPM	hp	kW
800	105	78
1,000	171	128
1,200	228	170
1,300	248	185
1,400	273	204
1,500	300	224
1,600	309	230
1,800	331	247
2,000	345	257
2,100	333	248



Fuel Consumption

RPM	lb/hp-hr	g/kW-hr
1,500	0.362	220
1,600	0.351	214
1,700	0.357	217
1,800	0.362	220
1,900	0.359	218
2,000	0.364	221
2,100	0.369	224

Curves shown above represent gross engine performance capabilities obtained and corrected in accordance with SAE J1995 conditions of 29.61 in Hg (100 kPa) barometric pressure [300ft (91m) altitude] 77 deg F (25 deg C) inlet air temperature, and 0.30 in Hg (1kPa) water vapor pressure with No. 2 diesel fuel. At speeds above 1,100 RPM the engine may be operated at altitudes up to 9,000 ft (2,743 m) before electronic derate is applied. At engine speeds below 1,100 RPM the engine may be operated at altitudes up to 970 ft (296 m) before electronic derate is applied.

STATUS FOR CURVES AND DATA: Final-(Measured data)

TOLERANCE: Within +/- 5

CHIEF ENGINEER:

J David Dixon

Intake Air System

Maximum allowable air temperature rise over ambient at Intake Manifold (Naturally Aspirated Engines) or Turbo Compressor inlet (Turbo-charged Engines): (This parameter impacts emissions, LAT and/or altitude capability)	30 delta deg F	16.7 delta deg C

Cooling System

Maximum charge air cooler outlet to ambient at 25 deg C [77 deg F] (CAC dT)	63.0 delta deg F	35.0 delta deg C
Maximum CAC outlet temperature at <=25 °C (77 °F) ambient	140 deg F	60 deg C
Maximum allowable pressure drop across charge air cooler and OEM CAC piping (IMPD):	4 in-Hg	13.5 kPa
Charge air cooler outlet temperature for full Fan-On	140 deg F	60 deg C
Maximum coolant temperature for engine protection controls	225 deg F	107 deg C
Maximum coolant operating temperature at engine outlet (max. top tank temp):	225 deg F	107 deg C

Exhaust System

Maximum exhaust back pressure:	10 in-Hg	34 kPa
Recommended exhaust piping size (inner diameter):	0 in	0 mm

Lubrication System

Nominal operating oil pressure		
@ minimum low idle	10 psi	69 kPa
@ maximum rated speed	55 psi	379 kPa
Minimum engine oil pressure for engine protection devices		
@ minimum low idle	10 psi	69 kPa

Fuel System

Fuel cooling requirements (with diesel fuel)		
Maximum heat rejection to return fuel at max. coolant and inlet fuel temperature:	52 BTU/min	0.91 kW
@ fuel return flow rate of:	115 lb/hr	52 kg/hr
@ fuel return temperature prior to cooler:	170 deg F	77 deg C
Maximum supply fuel flow:	277 lb/hr	126 kg/hr
Maximum return fuel flow:	137 lb/hr	62 kg/hr
Engine fuel compatibility (consult Service Bulletin #3379001 for appropriate use of other fuels)	ULSD, B5, B20	
Maximum fuel inlet pressure:	10 psi	69 kPa

Performance Data

Maximum low idle speed:	1,200 RPM
Minimum low idle speed:	700 RPM
Minimum engine speed for full load sustained operation:	1,500 RPM

	Rated Power		Maximum Power		Torque Peak	
Engine Speed	2,100 RPM		2,000 RPM		1,500 RPM	
Output Power	333 hp	248 kW	345 hp	257 kW	300 hp	224 kW
Torque	833 lb-ft	1,129 N-m	906 lb-ft	1,228 N-m	1,050 lb-ft	1,424 N-m
Friction Horsepower	87 hp	65 kW	76 hp	57 kW	44 hp	33 kW
Intake Manifold Pressure	59 in-Hg	200 kPa	58 in-Hg	196 kPa	67 in-Hg	225 kPa
Turbo Comp. Outlet Pressure	65 in-Hg	219 kPa	63 in-Hg	214 kPa	70 in-Hg	236 kPa
Turbo Comp. Outlet Temperature	370 deg F	188 deg C	365 deg F	185 deg C	400 deg F	204 deg C
Inlet Air Flow	647 ft3/min	305 L/s	620 ft3/min	293 L/s	515 ft3/min	243 L/s
Charge Air Flow	46.7 lb/min	21.2 kg/min	45 lb/min	20 kg/min	37 lb/min	17 kg/min
Exhaust Gas Flow	1,263 ft3/min	596 L/s	1,276 ft3/min	602 L/s	1,078 ft3/min	509 L/s
Exhaust Gas Temperature	905 deg F	485 deg C	962 deg F	517 deg C	862 deg F	461 deg C
Maximum Fuel Flow to Pump	277 lb/hr	126 kg/hr				
Heat Rejection to Coolant	8,454 BTU/min	149 kW	8,419 BTU/min	148 kW	7,346 BTU/min	129 kW
Heat Rejection to Fuel						
Heat Rejection to Ambient	1,096 BTU/min	19 kW	1,357 BTU/min	24 kW	1,612 BTU/min	28 kW
Heat Rejection to Exhaust	10,646 BTU/min	187 kW	11,010 BTU/min	194 kW	8,062 BTU/min	142 kW

**When operating Naturally Aspirated engines above SAE J1995 conditions, it should be noted that smoke levels will increase due to combustion inefficiencies associated with a reduction in the air to fuel mixture.

Cranking System (Cold Starting Capability)

Unaided Cold Start:				
Minimum cranking speed	120	RPM		
Minimum ambient temperature for unaided cold start	-10	deg F	-23.3	deg C
Cranking torque at minimum unaided cold start temperature	280	lb-ft	380	N-m
Aided Cold Start:				
Minimum ambient temperature with Grid Heater only	-10	deg F	-23	deg C
Minimum ambient temperature with coolant and lube heater only	-40	deg F	-40	deg C
Cold starting aids available	Intake Manifold Heater, Block Heater, Oil Pan Heater			
Maximum parasitic load at 10 deg F @ 125	308	lb-ft	418	N-m

Noise Emissions

Top	95.7	dBa
Right Side	97	dBa
Left Side	99.3	dBa
Front	99.6	dBa
Exhaust noise emissions	103.3	dBa

Estimated Free Field Sound Pressure Level at 3.28ft (1m) and Full-Load Governed Speed
(Excludes Noise from Intake, Exhaust, Cooling System and Driven Components)

Note: Maximum noise level.

Change Log

Date	Author	Change Description
6/29/2012	Kyle Roshak	Initial release of preliminary datasheet
7/1/2012	Kyle Roshak	Updated to alpha-estimated level
7/19/2012	Kyle Roshak	Added data
5/1/2013	Kyle Roshak	Added Fuel Consumption table.
5/6/2013	Kyle Roshak	Changed maximum exhaust backpressure from 10.1 in Hg to 10.0 in Hg.
5/16/2013	Kyle Roshak	Changed Heat Rejection to Coolant at peak torque from 7218 to 7972 BTU/min. Changed Charge Air Flow at peak torque from 36.4 to 39.3 lb/min.
6/3/2013	Kyle Roshak	Added torque curve at 5500 ft.
6/20/2013	Kyle Roshak	Changed to Beta confidence level. Added Motoring Friction Horsepower.
10/24/2013	Kyle Roshak	Updated to Final Measured status. Updated Performance Data Chart. Added Noise data. Updated Fuel Consumption curve. Updated Emission Certifications from pending to certified.
3/10/2014	Kyle Roshak	Corrected NTE zone. Updated Noise data.
5/5/2015	Kyle Roshak	Corrected compression ratio.

End of Report